Milestone 2: Machine Learning Model Development for Exoplanet Habitability Prediction

1. Introduction

- The goal of this milestone is to develop a machine learning model capable of predicting the potential habitability of exoplanets using astrophysical features. With the increasing number of discovered exoplanets, automated systems are necessary to identify planets that may support life.

- Machine learning models can analyse complex relationships between planetary characteristics such as mass, radius, orbital parameters, stellar properties, and equilibrium temperature to estimate the likelihood that a planet could be habitable.

- In this project, supervised learning techniques were used to classify planets into habitable or non-habitable categories and assign a probability score representing their habitability potential.

2. Data Preprocessing

Data preprocessing is a critical step in machine learning as raw astronomical datasets often contain inconsistencies, missing values, and features with different scales.

The preprocessing pipeline ensures that the dataset is clean, structured, and suitable for machine learning models.

2.1 Handling Missing Values

Astronomical datasets often contain missing values due to observational limitations or incomplete measurements.

To handle missing data, appropriate imputation strategies were applied:

- Numerical features were filled using statistical measures such as **mean or median values**.

- Missing categorical values were replaced with a placeholder category such as **"Unknown"**.

Handling missing values prevents model errors and ensures that all observations remain usable for training.

2.2 Feature Selection

The dataset contains several planetary and stellar properties that influence habitability.

Important features include:

- Planetary radius
- Planetary mass
- Orbital period
- Semi-major axis
- Equilibrium temperature
- Stellar temperature
- Stellar luminosity
- Stellar metallicity
- Stellar spectral type

These features were selected because they have a direct scientific relationship with planetary habitability.

For example:

- Equilibrium temperature determines whether liquid water may exist.
- Stellar luminosity affects the habitable zone of the star.
- Orbital stability influences long-term planetary climate conditions.

2.3 Feature Scaling

Many machine learning algorithms perform better when features are on similar numerical scales.

Since planetary attributes vary significantly in magnitude, **StandardScaler** was used to normalize the numerical features.

Standard scaling transforms data to have:

- Mean = 0
- Standard deviation = 1

This prevents features with large values from dominating the model training process.

2.4 Feature Encoding

Categorical features such as stellar spectral type cannot be directly used in machine learning algorithms.

To convert these into numerical representations, **One-Hot Encoding** was applied.

For example:

Star Type	Encoded Representation
G	[0,1,0,0,0]
K	[0,0,1,0,0]

This transformation allows the model to interpret categorical information effectively.

3. Machine Learning Model Training

After preprocessing the dataset, machine learning models were trained to classify exoplanets based on their habitability potential.

Two primary models were implemented:

- Random Forest Classifier
- XGBoost Classifier

These models are particularly effective for structured tabular datasets.

3.1 Random Forest Classifier

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and combines their predictions.

Advantages of Random Forest include:

- Ability to capture non-linear relationships
- Robustness to noise and outliers
- Reduced risk of overfitting
- Capability to estimate feature importance

Each decision tree in the forest is trained using a random subset of features and training samples. The final prediction is obtained through majority voting among all trees.

3.2 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a highly optimized implementation of gradient boosting designed for high performance and efficiency.

XGBoost improves prediction accuracy by sequentially building decision trees that correct the errors of previous trees.

Advantages of XGBoost include:

- High predictive performance
- Ability to capture complex feature interactions
- Built-in regularization to reduce overfitting
- Efficient computation for large datasets

Due to its strong performance on tabular data, XGBoost is widely used in data science competitions and real-world predictive systems.

4. Model Evaluation

To evaluate the performance of the trained models, several classification metrics were used.

4.1 Accuracy

Accuracy measures the proportion of correct predictions made by the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy provides an overall performance measure, it may not be sufficient for imbalanced datasets.

4.2 Precision

Precision evaluates the reliability of positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

High precision means that when the model predicts a planet as habitable, it is likely to be correct.

4.3 Recall

Recall measures the model's ability to detect actual habitable planets.

$$Recall = \frac{TP}{TP + FN}$$

High recall ensures that fewer habitable planets are missed during classification.

4.4 F1 Score

The F1 score provides a balance between precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

This metric is particularly useful when the dataset is imbalanced.

4.5 ROC-AUC Score

The ROC-AUC metric evaluates the model's ability to distinguish between habitable and non-habitable planets based on predicted probabilities.

Higher ROC-AUC values indicate better classification performance.

5. Hyperparameter Tuning

To improve model performance, hyperparameter tuning was performed using RandomizedSearchCV.

This technique tests multiple parameter combinations and selects the configuration that produces the best evaluation score.

Example parameters tuned:

Random Forest

- Number of trees (n_estimators)
- Maximum tree depth (max_depth)
- Minimum samples required for splitting

XGBoost

- Learning rate
- Maximum tree depth
- Number of boosting rounds

Hyperparameter tuning helps improve model accuracy while preventing overfitting.

6. Habitability Scoring and Planet Ranking

Instead of providing only binary predictions, the trained model outputs probability scores indicating the likelihood of habitability.

Using these probabilities:

1. Each planet receives a habitability probability score.
2. Planets are ranked from most habitable to least habitable.
3. The ranking is saved to a CSV file for analysis.

This ranking system helps prioritize candidate planets for further astronomical study.

7. Model Interpretability

Understanding which features influence the model's predictions is essential in scientific applications.

Feature importance analysis was used to determine which planetary characteristics contribute most to habitability predictions.

Common influential features include:

- Equilibrium temperature
- Stellar luminosity
- Orbital stability
- Planet radius
- Semi-major axis

These factors align with astrophysical theories regarding planetary habitability

9. Model Comparison and Final Model Selection

Two machine learning models were trained and evaluated for the habitability prediction task:

- Random Forest Classifier
- XGBoost Classifier

Both models were evaluated using several performance metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC score.

Among these metrics, F1-score and Recall were given higher importance, since correctly identifying potentially habitable planets is critical. Missing a truly habitable planet would be a significant scientific loss, therefore models with higher recall were preferred.

After evaluation, the XGBoost model demonstrated superior performance compared to Random Forest in terms of:

- Higher F1-score
- Better recall for habitable planets
- More stable performance on validation data
- Better ability to capture complex nonlinear relationships in astrophysical data

XGBoost also provides regularization mechanisms, which help reduce overfitting and improve generalization on unseen data.

Due to these advantages, XGBoost was selected as the final model for the habitability prediction system.

The trained XGBoost model was then saved as a serialized .pkl file and used to generate habitability probability scores for all exoplanets in the dataset.

Model Interpretability & Justification

1. Executive summary

- **Final model chosen:** XGBoost classifier (selected based on higher F1 and recall and stable validation performance).
- **Interpretability goals:** show which features drive predictions, explain the top contributors, and provide scientific reasoning linking model signals to planet habitability.
- **Deliverables in this section:** feature-importance plot, per-feature explanations, scientific rationale, recommended figures to include

2. Feature importance (visual + how to produce)

- **What to include:** a sorted bar plot of feature importances (descending) labelled with feature names and importance values.
- **Recommended filename & placement:** figures/feature_importance.png — insert immediately after this subsection with caption *Figure 1: Feature importance (XGBoost)*.
- **Brief code :**

```
# assuming `final_model` is the trained XGBoost pipeline and `feature_names` list is available

import matplotlib.pyplot as plt

import numpy as np


# if pipeline: extract estimator and feature names appropriately

est = final_model.named_steps.get('clf', final_model) # adjust key if different

importances = est.feature_importances_

idx = np.argsort(importances)[::-1]
```

```
plt.figure(figsize=(10,6))
plt.barh([feature_names[i] for i in idx], importances[idx])
plt.gca().invert_yaxis()
plt.title('Figure 1: Feature importance — XGBoost')
plt.xlabel('Importance')
plt.tight_layout()
plt.savefig('figures/feature_importance.png', dpi=200)
plt.show()
```

3. Top contributing features — list & short plain explanation

The list below assumes the model ranked these features highest (replace with your actual top N features from the plot).

1. **Equilibrium temperature (eq_temp_k)**
   - *Model role:* strongest positive/negative signal for habitability probability.
   - *Why scientifically relevant:* surface temperature determines whether liquid water can exist; planets with equilibrium temperatures near Earth-like ranges are more likely to be habitable.
2. **Radius / radius_earth or pl_rade**
   - *Model role:* helps separate rocky terrestrial planets from gas giants (smaller radius → higher habitability probability in many cases).
   - *Why scientifically relevant:* planets larger than ~1.5–2.0 $R_\oplus$ tend to retain thick H/He envelopes, decreasing surface habitability prospects.
3. **Semi-major axis / Orbital period (semimajor_axis / orbital_period)**

- *Model role:* determines incident flux and seasonal stability.
- *Why scientifically relevant:* a planet's distance from its star defines the habitable zone position; orbital stability affects long-term climate.

4. **Host star effective temperature (star_temp_k) / Spectral type flags**
    - *Model role:* moderates habitable zone boundaries and stellar activity proxies.
    - *Why scientifically relevant:* star type and temperature set the radiation environment, spectral energy distribution, and lifetime — all critical for habitability windows.
5. **Planet mass (mass_earth) or density**
    - *Model role:* influences surface gravity and atmosphere retention (both shape habitability).
    - *Why scientifically relevant:* sufficient mass supports atmosphere and plate tectonics; too massive may prevent Earth-like conditions.
6. **Stellar luminosity (star_luminosity)**
    - *Model role:* combined with semi-major axis gives incident flux; strong predictor of thermal environment.
    - *Why scientifically relevant:* determines energy input to the planet and habitable zone location.
7. **Orbital/stellar stability metrics (orbital_stability, stellar_compatibility)**
    - *Model role:* penalises planets in highly eccentric/unstable or multi-star unstable orbits.
    - *Why scientifically relevant:* chaotic or highly eccentric orbits cause extreme climate swings, limiting stability for life.

4. Scientific reasoning behind model signals (short bullets)

- **Liquid water first principle:** features that control surface temperature and pressure (stellar luminosity, semi-major axis, equilibrium temperature) dominate because liquid water is the primary habitability proxy used in the label definition.
- **Rocky vs gaseous distinction:** radius and mass separate terrestrial planets from gas giants; habitability probability increases for smaller radii when other conditions are favorable.
- **Host star influence:** spectral type and stellar temperature matter for both habitable zone geometry and high-energy radiation, which affects atmosphere retention and surface chemistry.
- **Stability matters:** long-term orbital and stellar stability are necessary for sustained biospheres; the model captures this through orbital stability features.
- **Model caution:** model correlations are not proof of causation — some features may correlate with observational biases (e.g., detection method → biases in measured radius/period). Discussed below.

5. Interpretability methods recommended

- **Global importance:** use XGBoost built-in feature_importances_ and SHAP summary plot for consistent ranking.
- **Local explanations:** use SHAP force plots for a few top candidate planets to show why the model scored them highly.
- **Partial dependence plots:** for eq_temp_k and radius_earth to visualize marginal effects.

Conclusion

This milestone implemented a complete machine learning pipeline for predicting exoplanet habitability. The process involved data preprocessing, feature engineering, model training, hyperparameter tuning, and performance evaluation.

Two models, Random Forest and XGBoost, were trained and evaluated using multiple classification metrics. The final system generates probability-based habitability scores and ranks exoplanets accordingly.

The developed model provides an automated framework that can assist astronomers in identifying promising candidates for habitable planets and supports future research in astrobiology and planetary science.